1. Clinical Question & Core Hypothesis

Clinical Question

In men on active surveillance some apparently low-risk prostate tumors progress — why do tumors that acquire specific chromatin states under oncogenic MUC1-C activity escape surveillance and progress? 2) Tier 1 path used: the single mechanistic chain MUC1-C → regulation of accessible chromatin in tumor cells → phenotypic consequences (tumor biology and ctDNA signals), plus the PTEN axis linking recurrence/progression to altered localization/signaling. I built on MUC1-C as the chromatin regulator and PTEN as the key tumor suppressor whose localization/function modulates downstream signaling. 3) Tier 2/supporting evidence (modalities): scATAC-seq (measures MUC1-C–driven accessible chromatin states) and ctDNA/nomogram logic (translate cell-intrinsic chromatin changes into blood-measurable prognostic signals usable in clinical risk models). These support a translational path from single-cell chromatin to circulating biomarkers and decision tools. 4) Endpoint choice (PFI): Active surveillance decisions hinge on delayed progression (time to need for treatment); progression-free interval directly captures the clinically relevant event for surveillance management. 5) Deliberate exclusions: I excluded broad multi-gene panels, immune-imaging features, and complex microenvironmental gene lists to keep the hypothesis mechanistically sharp (max 1 imaging feature + 2 genes). I did not introduce genes not present in the path (e.g., AKT/PIK3CA) and left other modalities (nomogram implementation details) to downstream work?

Working Hypothesis

MUC1-C–driven remodeling of chromatin in prostate tumor cells suppresses PTEN localization and function, promoting transcriptional programs that drive stromal remodeling and yield a high stromal-to-tumor ratio (TSR). Consequently, tumors with high TSR together with evidence of MUC1-C activation and PTEN dysregulation will exhibit shorter progression-free intervals on active surveillance, and integrating ctDNA-based MUC1-C chromatin signatures alongside scATAC-derived chromatin features into prognostic nomograms will improve PFI prediction.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MUC1, PTEN
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MUC1: MUC1-C activity escape surveillance and progress? 2) Tier 1 path used: the single mechanistic chain MUC1-C → regulation of accessible chromatin in tumor cells → phenotypic consequences (tumor biology and ctDNA signals), plus the PTEN axis linking recurrence/progression to altered localization/signaling.

PTEN: PTEN as the key tumor suppressor whose localization/function modulates downstream signaling.

3. Clinically Testable Predictions

Prediction 1: Assemble a retrospective cohort of men on active surveillance with available tumor tissue and longitudinal follow-up; quantify TSR via digital pathology, and assess MUC1-C expression/localization and PTEN localization/signaling (p-AKT as a readout) by IHC.

Prediction 2: Perform scATAC-seq on tumor samples to identify chromatin accessibility changes associated with MUC1-C activity and link these patterns to TSR status.

Prediction 3: Develop a ctDNA assay capturing MUC1-C–associated chromatin signatures (e.g., methylation or fragmentomics proxies for MUC1-C target regions) and test their association with PFI.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions